

Parent's Signature:

CANDIDATE'S NAME: _____ **CTG:** _____

YISHUN JUNIOR COLLEGE
2017 JC2 PRELIMINARY EXAMINATION

CHEMISTRY HIGHER 2

9729/02

Paper 2 Structured Questions

THURSDAY 24 AUGUST 2017

1400 hrs – 1600hrs

(2 hours)

Candidates answer on the Question Paper

Additional Materials: Data Booklet



READ THESE INSTRUCTIONS FIRST

Write your name and CTG on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 1	
Total	/ 30
Paper 2	
1	/ 15
2	/ 12
3	/ 10
4	/ 19
5	/ 19
Total	/ 75
Paper 3	
Total	/ 80
Paper 4	
Total	/ 55

Answer **all** the questions.

- 1** Antacids neutralise stomach acid and relieve the symptoms of indigestion, heartburn or stomach ulcer. One of the active ingredient in antacid is calcium carbonate, CaCO_3 . There are also other ingredients such as binders present in each tablet. On average, a 1.3 g tablet contains 0.5 g of CaCO_3 .

Calcium carbonate undergoes neutralisation reaction with hydrochloric acid as shown in the following equation.



A student was given an antacid tablet. She was asked to determine the volume of stomach acid that can be neutralised by 1 tablet. She crushed the antacid tablet and dissolved it with 50.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ of HCl in a 50.0 cm^3 graduated flask. She pipetted 25.0 cm^3 of this sample and boiled gently for 5 minutes. She then titrated the boiled sample with 20.0 cm^3 of $0.0500 \text{ mol dm}^{-3}$ sodium hydroxide for complete neutralisation.

- (a)** Suggest a reason why the 25.0 cm^3 sample was boiled before titrating with NaOH .

.....[1]

- (b) (i)** Calculate the amount of HCl present initially in the graduated flask.

[1]

- (ii)** Calculate the amount of HCl reacted with NaOH .

[1]

- (iii) Using your answer in **(b)(i)** and **(b)(ii)**, calculate the amount of HCl neutralised by one antacid tablet.

[2]

- (iv) A patient with a hyperacidic stomach has a HCl concentration of $0.0300 \text{ mol dm}^{-3}$. The volume of liquid in the stomach is 375 cm^3 .

Using your answer in **(b)(iii)**, calculate the minimum number of antacid tablets that the patient with hyperacidic stomach would have to take to bring the concentration of HCl in the stomach to the normal level of $0.000300 \text{ mol dm}^{-3}$.

[2]

- (c) The following diagram shows the nutrition fact label of the antacid used for the student's analysis.



Verify by calculation if there are 500 mg of CaCO_3 in each antacid tablet.

[2]

- (d) Magnesium oxide, MgO and aluminium oxide, Al_2O_3 can also act as antacids. The oxides MgO and Al_2O_3 have the melting points 2852°C and 2072°C respectively.
- (i) Both MgO and Al_2O_3 have giant ionic structure. Explain the difference between the melting points of MgO and Al_2O_3 .

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.....

.....[2]

- (ii) Describe the reactions of MgO and Al_2O_3 with water. Give the approximate pH of any solution formed.

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.....
.....[2]

- (iii) When NaOH(aq) is added to separate solutions containing $\text{Mg}^{2+}(\text{aq})$ and $\text{Al}^{3+}(\text{aq})$, a white precipitate is formed in each case. When an excess of NaOH(aq) is added, **one** of these precipitates dissolves.

Construct ionic equations, with state symbols, for the formation of the white precipitate, and its subsequent dissolving.

.....
.....
.....[2]

[Total: 15]

- 2 The halogens, found in group 17 of the Periodic Table, and their compounds are useful laboratory reagents.

The table below shows some properties of the halogens and their compounds.

halogen	chlorine	bromine	iodine
standard enthalpy change of vapourisation ($\Delta H_v^\ominus$) of X_2 / kJ mol^{-1}	0	+15	+30
electronegativity	3.2	3.0	2.7
standard enthalpy change of reaction ($\Delta H_r^\ominus$) for thermal decomposition of HX / kJ mol^{-1}	+183	+103	+11
solubility of $\text{PbX}_2(\text{s})$ / g dm^{-3}	4.7	4.3	0.6

- (a) Describe and explain the trend in the volatility of the halogens.

.....

[2]

- (b) By quoting relevant data from the *Data Booklet*, explain the relative thermal stabilities of the hydrides of the Group 17 elements.

.....

[2]

- (c) (i) Write an expression for K_{sp} of PbCl_2 , stating the units.

$K_{\text{sp}} = \dots\dots\dots \text{units} \dots\dots\dots$ [1]

- (ii) Using relevant data from the table and your expression for K_{sp} in (c)(i), calculate the solubility product of $PbCl_2$.

Give your answer to **two** significant figures.

[2]

- (d) Astatine, At, is found below iodine in group 17 of the Periodic Table. Using your knowledge of the chemistry of group 17, predict the changes (if any) that you may observe when the following reagents are mixed.

If nothing is formed, write “no reaction” and where there is a reaction predicted, write an ionic equation with state symbols.

- (i) aqueous bromine and aqueous sodium astatide

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.....
.....
.....[1]

- (ii) astatine and aqueous sodium chloride

.....
.....
.....
.....[1]

- (e) In 1886, Henri Moissan succeeded in obtaining fluorine by the electrolysis of molten potassium hydrogen difluoride, KHF_2 , which is an ionic compound containing **one** cation and **one** anion.

- (i) Write the formulae of the ions present in KHF_2 .

cation:

anion:

[1]

- (ii) Draw the structure for the anion and indicate the types of bonding that occur within it in your diagram.

[2]

[Total: 12]

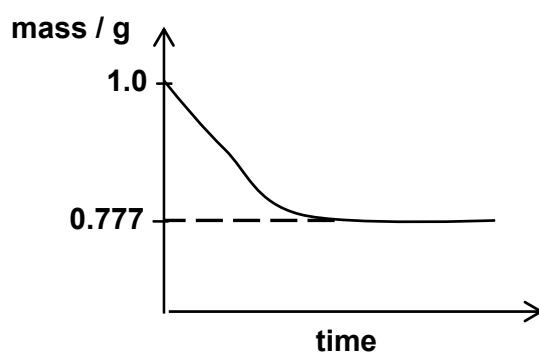
- 3 (a) (i) Describe and explain how the thermal stability of magnesium carbonate might compare to calcium carbonate.

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.....
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.....
.....
.....[3]

- (ii) Write an equation for the decomposition of MgCO_3 .

.....[1]

- (b) The graph represents the change in mass that occurs when 1.0 g of powdered carbonate of another group 2 element, **Z**, is decomposed at a temperature T.



Using the above graph, calculate and determine the identity of **Z**.

Identity of **Z** :

[3]

- (c) Beryllium, the first member of the group 2 elements was discovered in 1798 by a chemist named Vaquelin when he was working with emeralds. Cutting edge computers require an alloy of beryllium and copper to meet the demands placed on microprocessor connectors.
- (i) Beryllium chloride reacts with ammonia in a 1:2 ratio to form a product. Draw dot-and-cross diagrams showing the bonding in beryllium chloride and ammonia.



beryllium chloride



ammonia

[1]

- (ii) Draw a diagram to illustrate the shape of the product around the Be atom formed from the reaction between beryllium chloride and ammonia.

[2]

[Total: 10]

- 4** Energy density is the amount of energy that can be stored in a given mass of a substance or system. The higher the energy density of a substance or system, the greater the amount of energy stored in its mass. In fuel, energy density can be measured as the amount of energy released in megajoules per gram of fuel (MJ/g).
(1 MJ = 10^3 kJ)

The table below provides information on the standard enthalpy change of combustion of some commonly used fuel today.

fuel	formula	standard enthalpy change of combustion ($\Delta H_c^\ominus$) / kJ mol ⁻¹
hydrogen	H ₂	- 286
coal	C	- 393
ethanol	CH ₃ CH ₂ OH	- 1367
ethene	CH ₂ =CH ₂	- 1410

- (a)** An average household was found to consume an average energy of 108 MJ per day.
- (i)** Using relevant data from the table above, determine the mass of the coal required to supply energy for an average household a day.

Hence, determine the energy density of coal.

[2]

- (ii)** The energy density of ethanol was found to be 0.0297 MJ/g. With reference to your answer in **(a)(i)**, state and explain which fuel is more efficient in supplying energy to the average household.

.....

.....

.....

.....

.....[2]

- (iii) Ethanol, ethene and liquid hydrogen are possible alternative sources of fuel that may be used in replacement of coal as the burning of coal releases toxic carbon monoxide gas into the atmosphere.

Explain why carbon monoxide gas, CO, is toxic.

.....
.....
.....
.....[2]

- (b) (i) Using relevant data from the table, determine the standard enthalpy change of formation of liquid ethanol.

[1]

- (ii) Using relevant data from the *Data Booklet*, determine the standard enthalpy change of combustion of liquid ethanol.

[2]

- (iii) Suggest a reason for the discrepancy between the value in (b)(ii) and the one given in the table.

.....
[1]

- (c) The direct oxidation of alcohols such as ethanol in a fuel cell represents potentially the most efficient method of obtaining useful energy from a renewable fuel.

- (i) Given that carbon dioxide is formed at the anode while water is produced at the cathode, construct the relevant half-equations for the reactions occurring at the electrodes and hence the equation for the overall reaction, assuming acidic conditions.

cathode:

anode:

overall:

[2]

- (ii) The value for the E° for the reaction occurring at the anode was determined to be +0.08 V. Using suitable data from the *Data Booklet*, determine the e.m.f produced by the ethanol fuel cell.

[2]

- (iii) Hence, determine the value of ΔG° of the ethanol fuel cell.

[1]

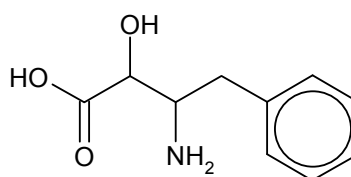
-[1]

- Type of hybridization in C:

.....[3]

[Total: 19]

- 5 (a) Compound **Y** is an intermediate in the production of a drug which is being developed to reduce the effects of the HIV virus.



compound **Y**

- (i) Name three functional groups in compound **Y**, other than the phenyl group.

.....
[2]

- (ii) On the structure above, indicate with an asterisk (*) any chiral carbon present, and state the total possible number of stereoisomers that compound **Y** can have.

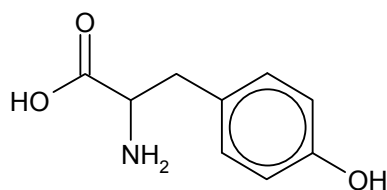
total possible number of stereoisomers = [2]

- (iii) Draw the structural formula of the organic products that will be formed when compound **Y** is treated with

- hot acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$

- anhydrous $\text{PBr}_3(l)$

- (b) Compound **Y** is similar in structure to tyrosine, one of the twenty α -amino acids found in proteins.



tyrosine

At pH = 5.7, compound **Y** exists as a *zwitterion*.

- (i) Define the term *zwitterion*.

.....
[1]

- (ii) Draw the structural formula of the zwitterionic form of tyrosine.

[1]

Another potential anti-HIV drug can be synthesised by combining tyrosine with another α -amino acid, glycine, $\text{H}_2\text{NCH}_2\text{CO}_2\text{H}$.

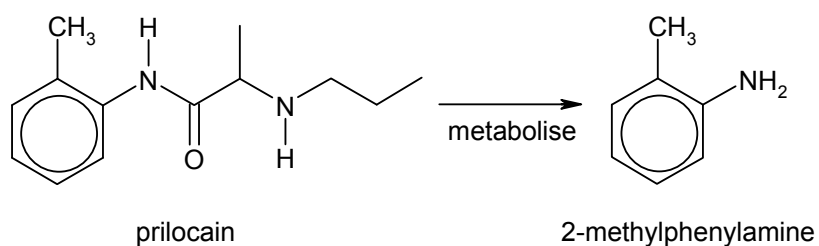
- (iii) State the type of linkage that will be formed between tyrosine and glycine.

.....[1]

- (iv) Draw the structures of two different dipeptides that could be formed when one molecule of tyrosine reacts with one molecule of glycine.

[2]

- (c)** Prilocain, a widely used local anaesthetic, is metabolised in the liver by enzymes to form 2-methylphenylamine.



- (i) State the reagents and conditions required to produce 2-methylphenylamine from prilocain in the laboratory.

.....[1]

In the aqueous medium, 2-methylphenylamine is a weak Bronsted base.

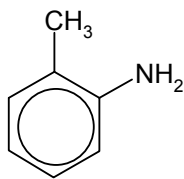
- (ii) Write an expression for the base dissociation constant, K_b , for 2-methylphenylamine.

[1]

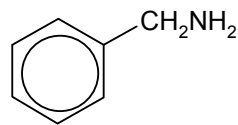
- (iii) The pK_b value for 2-methylphenylamine is 9.6. Use this value to calculate the pH of a 0.10 mol dm^{-3} solution of 2-methylphenylamine.

[2]

- (d) 2-methylphenylamine and benzylamine are isomers.



2-methylphenylamine



benzylamine

Suggest how the pK_b value of benzylamine might compare to that of 2-methylphenylamine.

Explain your answer in terms of their structures.

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.....[3]

[Total: 19]

~ END OF PAPER ~